

F24-108-D-PurrfectAssistant

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Chapter 1

Introduction

Purrfect Assistant is a comprehensive mobile application for cat care and this document outlines this app's requirements. Features of the app include diet tracking, breed detection, health monitoring, and a GPS tracker for real time cat location. It is meant for developers to help guide implementation, testers to validate functionality, and users looking for a custom cat care solution. This document will also help project managers and documentation writers to plan and maintain the project.

1.1 Problem Statement

Scattered and unreliable resources make it difficult for many cat owners to manage their pet's health, diet and overall care. As a result, you get inconsistent feeding, missed health warnings and poor training or exercise routines, often for new cat owners who are not familiar with their pet's special requirements. Another complicating factor is that it's difficult to identify specific breeds, and then understand their care requirements, so that you can provide personalized attention. Additionally, existing apps usually provide disjointed solutions for diet tracking, health monitoring and breed identification. This causes frustration to the users. Most apps aren't able to provide special care for cats with special care needs, such as newborns, rescued cats, and pregnant cats. Moreover, tracking a cat's location, when they go outdoors, is difficult and worrying for their owners. These gaps lead to pet stress, health problems and owner dissatisfaction, those who want better care for their pets. The Purrfect Assistant app is aiming to give a personalized, all in one solution to help cat owners. It will help their cats live a healthy and happy life.

1.2 Scope

The Purrfect Assistant project is a holistic solution to cat care that will help owners to easily manage and optimize their pets' health and wellbeing. This application is meant to help users keep track of their cat's diet, health, behavior and daily activities, with personalized recommendations and real time updates to improve their overall care experience. Through artificial intelligence, the app will use personalized data of your own cat to deliver personalized suggestions just for you. This will significantly improve health outcomes and foster a much closer relationship between a cat and its owner. This mobile application is aimed at cat owners who look for a comprehensive solution for daily care and health of their cat, easy and convenient for both the novice and experienced pet owners, with a focus on user friendliness and accuracy of data, as well as habit formation and behavioral tracking, so that cat owners can better take care of their cats. Purrfect Assistant app has the main aim to be the best tool for cat owners, aiding in proactive management of their pets' health and well-being in a convenient way, so much so that the owners feel it becoming a necessity for them.

1.3 Modules

The project is divided into 6 main modules and then into further sub-features.

1.3.1 Breed Detection

This module will allow the user to upload a picture of their cat and the app will identify its breed.

1. Builtin Camera Integration
2. Near Real time Breed Detection

1.3.2 Training and Care

This module will use generative AI and the cat's personalized data to give personalized recommendations regarding every aspect of cat care. It will also let the user track their progress regarding the care routines and plans.

1. Cat Care
 - (a) Training

- (b) Grooming

- (c) Exercise

2. Special Care

- (a) Newborn Cats

- (b) Pregnant Cats

- (c) Neutering

1.3.3 Health and Nutrition

This module focuses solely and in depth on the health and nutrition of cats. It allows user to get personalized nutrition plans and follow and track these plans. It also allows users to analyze their cat's symptoms using a Cat Doctor feature. The module will also provide web scraped veterinarian clinics.

- 1. Diet Tracking

- 2. Personalized Nutrition Plans

- 3. Cat Doctor

- 4. Find Veterinarians

1.3.4 Accessories

This module provides users with tons of online cat accessories using web scraping. These are divided into categories such as litter boxes, toys and collars. Users will see various accessories with best prices. Sellers can also sponsor to promote their accessories.

- 1. Cat Accessory Finder

- 2. Price Filters

1.3.5 Cat Bot

This module will use an AI powered chat bot to answer questions and provide quick assistance to users who do not need a full-fledged plan. Instead they just need a short quick answer to a simple question regarding their cat's well-being.

- 1. Chat Bot

1.3.6 GPS Tracker

This module aims to improve cat safety when they are outdoors. It will provide a cheap and reliable solution to cat owners in scenarios where their cats get lost. A custom collar will be provided that will be configured in the app to track the cat's location.

1. Track Location
2. Order Collar

1.4 User Classes and Characteristics

User Classes and Characteristics

Table 1.1: User Classes and Their Descriptions

User class	Description
Cat Owners	Cat owners are individuals who have pet cats in their custody and the cats' well-being is their responsibility. They are their cats' primary care-givers. They can range from experienced owners to fairly new owners. They are the primary target of this app and every single module and feature is designed for their aid and benefit.
Veterinarians	They are medical professionals who specialize in animal care. They can use many of the features just like cat owners to get recommendation and assistance. They can monitor their clients progress on the app to get a better idea of the cat's personal preferences. They can also sponsor and market their own clinics on the app.
Cat Acces- sory Sellers	These are retailers and online sellers that specialize in cat care products such as litter boxes, cat toys and collars. They can benefit from this app by sponsoring and promoting their brand and products on the app.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

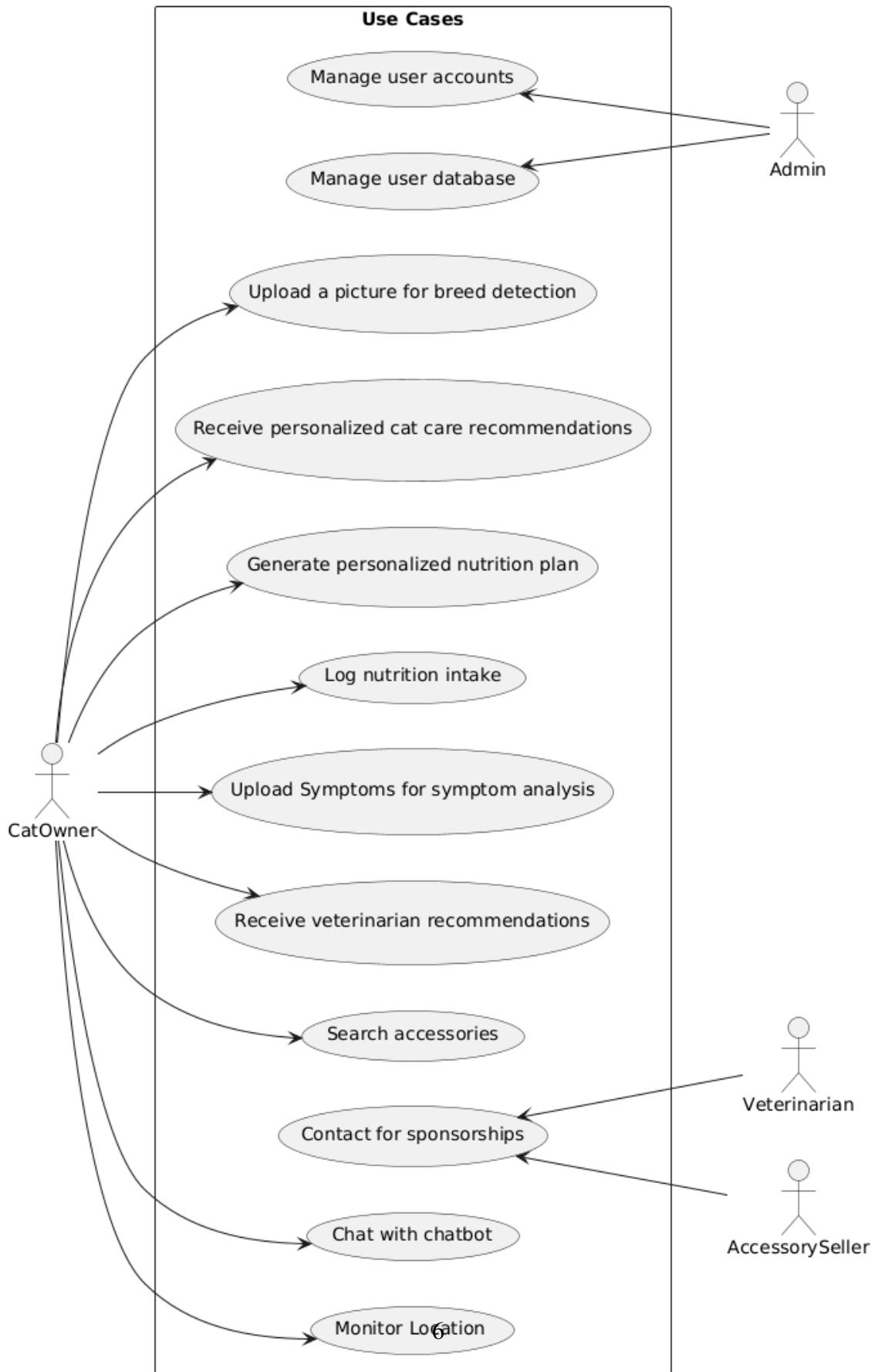


Figure 2.1: Use Case Diagram

2.1.1 UC1: Manage User Accounts

Actor: Admin

Description: In this use case the admin can manage user accounts within the Purrfect Assistant application. When necessary the admin can update account details and deactivate or delete accounts, and reset user passwords. We want to preserve the integrity of user data, and only allow authorized users to access the system.

Pre-condition: Admin should be logged into admin account with administrative privileges.

Post-condition: Accounts are successfully created, deleted and updated.

Main Flow:

- 1- The admin goes to the existing accounts option.
- 2- The admin selects a user account.
- 3- The admin selects delete account option.
- 4- The admin clicks confirm.
- 5- The system processes the deletion.

2.1.2 UC2: Manage User Database

Actor: Admin

Description: In this use case the admin is able to manage the entire user database of the Purrfect Assistant system. It allows the admin to see, search, backup, restore or export data of user in the database and to maintain its security and integrity. Database is maintained by the admin to store and control the access to the sensitive user information properly.

Pre-condition: Admin should be logged into admin account with administrative privileges.

Post-condition: Any changes to the user database are successfully made such as backup, export or modification.

Main Flow:

- 1- The admin goes to the existing accounts option.
- 2- The admin selects a user account.
- 3- The admin selects user data option.
- 4- The admin selects export data option.

5- The admin selects file type and confirms.

6- The system lets admin download the user data in specified file format.

2.1.3 UC3: Upload Picture for Breed Detection

Actor: User (Cat Owner)

Description: In this use case, users are able to upload a picture of their cat to the Purrfect Assistant app to get breed detection. The machine learning algorithms are used to process the image and identify the cat's breed in near real time that is returned to the user.

Pre-condition: The user should be logged in and camera permission should be given to the app.

Post-condition: The image is successfully uploaded and the system detects the breed accurately and shows it to the user.

Main Flow:

- 1- User navigates to breed detection category.
- 2- User clicks on upload image option.
- 3- User takes an image with their camera.
- 4- The system detects the breed of the cat.
- 5- The system displays the detected breed.

2.1.4 UC4: Receive Personalized Cat-Care Recommendations

Actor: User (Cat Owner)

Description: In this use case, users are able to get personalized cat care recommendations by their cat's data, such as age, breed, health conditions, etc. The cat specific system offers suggestions for training, grooming, exercise, and general well being.

Pre-condition: The user should be logged in.

Post-condition: The app successfully provides personalized recommendations regarding training, grooming, exercise and special care.

Main Flow:

- 1- User navigates to cat care option.
- 2- User clicks on training option.
- 3- User selects training type for example litter box training.

- 4- User fills out other details.
- 5- User clicks on Get Recommendation button.
- 6- System displays AI generated personalized recommendation for litter box training.

2.1.5 UC5: Generate Personalized Nutrition Plan

Actor: User (Cat Owner)

Description: This use case allows users to create a personalized cat nutrition plan for the cat based on cat age, weight, breed, health condition and dietary preferences. The system offers the cat's detailed meal recommendations, portion sizes and nutritional guidelines according to the cat's individual needs.

Pre-condition: The user should be logged in.

Post-condition: The app successfully generates personalized nutrition plan for cat.

Main Flow:

- 1- User navigates to cat diet category.
- 2- User clicks on generate nutrition plan.
- 3- User fills out necessary details.
- 4- User clicks on Generate Plan.
- 6- System displays AI generated personalized nutrition plan for cat.

2.1.6 UC6: Log Nutrition Intake

Actor: User (Cat Owner)

Description: In this use case users can log their cat's daily food intake, including type of meal, portion size, and feeding time.

Pre-condition: The user should be logged in.

Post-condition: The cat's nutrition intake is successfully logged and saved in the database.

Main Flow:

- 1- User navigates to cat diet category.
- 2- User clicks on Log Diet option.
- 3- User selects meal type.
- 4- User adds description.

5- User clicks on add.

6- System saves the logged meal.

2.1.7 UC7: Upload Symptoms for symptom analysis

Actor: User (Cat Owner)

Description: In this use case, the users can upload symptoms about their cat's health for analysis. It looks at the symptoms and gives user a potential diagnosis.

Pre-condition: The user should be logged in.

Post-condition: The symptoms are successfully uploaded and the app successfully provides a symptom diagnosis.

Main Flow:

- 1- User navigates to cat doctor option.
- 2- User clicks on symptom analysis option.
- 3- User fills out symptoms details.
- 4- User clicks on Diagnose button.
- 6- System displays AI generated diagnosis.

2.1.8 UC8: Receive Veterinarian Recommendations

Actor: User (Cat Owner)

Description: In this use case, the user will be shown nearby veterinarian locations and their contact info.

Pre-condition: The user should be logged in.

Post-condition: The app successfully provides veterinarian recommendations.

Main Flow:

- 1- User navigates to cat doctor option.
- 2- User clicks on Recommend Vets option.
- 3- System displays nearby vet clinics and contact information.

2.1.9 UC9: Search Accessories

Actor: User (Cat Owner)

Description:In this use case, we are allowing users to search for and browse through cat accessory available for purchase online. They are divided in categories like litter boxes, toys and collars and can be filtered by price.

Pre-condition: The user should be logged in.

Post-condition: The app successfully displays listings of cat accessories and products.

Main Flow:

- 1- User navigates to accessories option.
- 2- User clicks on Toys category.
- 3- User clicks on search.
- 4- System provides listings of Cat toys with descriptions and prices.

2.1.10 UC10: Contact for sponsor ships

Actors: Veterinarians, Accessory Sellers

Description:In this use case, vets and sellers can contact the app developers to sponsor and promote their clinics or products.

Pre-condition: The user should be logged in.

Post-condition: The vets and sellers successfully contact the developers.

Main Flow:

- 1- User navigates to contact for sponsorship option.
- 2- User clicks on either Vet or Accessory option.
- 3- User types in a message for why they are contacting and what they are offering.
- 4- User clicks send.
- 5- System sends data to developers.

2.1.11 UC11: Chat with chat bot

Actor: User (Cat Owner)

Description:In this use case, users can ask cat care related questions to a built in chat bot and get short quick answers.

Pre-condition: The user should be logged in.

Post-condition: The chat bot gives quick answers to the user.

Main Flow:

- 1- User navigates to cat bot option.
- 2- User types in a small cat care related question.
- 3- User clicks send.
- 4- System responds with appropriate answer.

2.1.12 UC12: Monitor Location

Actor: User (Cat Owner)

Description: In this use case, user will be able to locate their cat using a customized tracking collar and in built feature of the app.

Pre-condition: The user should be logged in and the cat collar is configured.

Post-condition: The app successfully shows the current location of the cat.

Main Flow:

- 1- User navigates to GPS tracker option.
- 2- User clicks on locate option.
- 3- System shows cat location on maps.

2.2 Functional Requirements

2.2.1 Breed Detection

Following are the functional requirements for Breed Detection module:

1. The app shall allow users to use the builtin phone camera to capture cat images within the app.
2. The app shall allow users to upload cat images to the app.
3. The app shall provide near real-time breed detection from the uploaded cat images.

2.2.2 Training and Care

Following are the functional requirements for Training and Care module:

1. The app shall allow user to generate a personalized cat training plan.

2. The app shall provide personalized grooming tips to the user.
3. The app shall allow user to generate a personalized exercise plan.
4. The app shall assist users to follow and track the training and exercise plans.
5. The app shall provide specialized guidance regarding newborn cats, pregnant cats and cat neutering.

2.2.3 Health and Nutrition

Following are the functional requirements for Health and Nutrition module:

1. The app shall allow users to generate personalized nutrition plans for their cats.
2. The app shall allow users to track their cat's daily nutrition intake.
3. The app shall allow users to input their cat's health issues and symptoms.
4. The app shall provide symptom analysis reports and further guidance.
5. The app shall provide nearest veterinarian clinic recommendations.
6. The app shall provide developer's contact info to veterinarians for potential sponsorships.

2.2.4 Accessories

Following are the functional requirements for Accessories module:

1. The app shall allow users to choose from different categories of accessories.
2. The app shall provide users with significant amount of products and listings.
3. The app shall allow users to set a price filter to get recommendations in that price range.
4. The app shall provide developer's contact info to sellers for potential sponsorships.

2.2.5 Cat Bot

Following are the functional requirements for Cat Bot module:

1. The app shall allow users to ask cat care related questions to an in built chat bot.
2. The app shall provide short and reliable answers to aid the users.

2.2.6 GPS Tracker

Following are the functional requirements for GPS Tracker module:

1. The app shall allow users to order a custom cat collar.
2. The app shall allow users to configure custom collar with the app.
3. The app shall provide real-time GPS location of cat collar.

2.3 Non-Functional Requirements

2.3.1 Reliability

REL-1: The app shall have a Mean Time Between Failures (MTBF) of at least 500 hours.

REL-2: The user data shall be backed up on back-end data base to avoid data loss.

2.3.2 Usability

USE-1: The app shall provide a simple and straight forward user interface.

USE-2: The app shall use input validation in chat bot to ensure cat related questions are catered.

USE-3: The app shall use input validation before any plan generation to ensure accurate plans.

USE-4: The system shall allow users to complete primary functions such as nutrition logging within 3 clicks or less.

2.3.3 Performance

PER-1: The app shall identify breed of uploaded images within 10 seconds.

PER-2: The app shall have an accuracy of at least 90% for correct breed detection.

PER-3: The app shall generate a personalized nutrition, training and exercise plans within 10 seconds.

PER-4: The symptom analysis should be accurate to at least 90%.

PER-5: The GPS tracker location should be accurate to within at-least 15 meters.

PER-6: The app shall be able to handle at least 500 concurrent users initially.

2.3.4 Security

SEC-1: The system shall manage user accounts with login IDs and passwords to keep user data and progress safe.

SEC-2: The login and sign-up system shall be resistant to common vulnerabilities such as SQL injections.

2.4 Domain Model

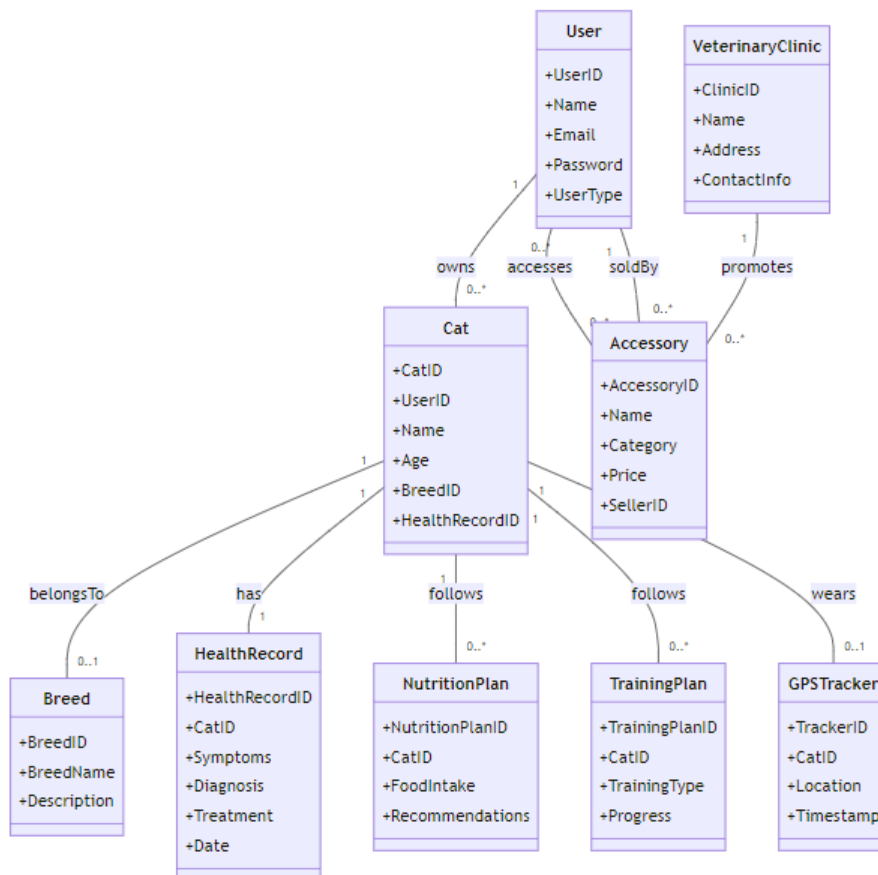


Figure 2.2: Domain Model

Chapter 3

System Overview

Purrfect Assistant is a mobile app that offers all in one care and management for cat owners. The app is designed to help cats to be healthier and more well, by providing users with tools to track their pet's diet, health, behaviour and daily activity. Purrfect Assistant is a more personalized way to connect cats with their owners through personalized recommendations and real time updates to make sure their care needs are met.

Artificial intelligence is used by the app to analyze personalized data to help users optimize cat health outcomes. Purrfect Assistant makes the app user friendly and easy to use for both novice and experienced pet owners, making it a need to have for proactive cat care management.

Background and Context: In the current times, pet ownership is increasing and cat owners are looking for more effective tools to manage their pets' health and wellbeing. However, many owners find it difficult to provide adequate care because there are no personalized resources and information that are specifically for their particular needs. This gap is addressed by Purrfect Assistant, an AI based solution which integrates modern technology and cat care management

Functionality: It is structured in several core modules such as Breed detection, Training and Care, Health and Nutrition, Accessories, a Chat Bot for immediate assistance and a GPS Tracker to track a cat's location. The modules are created with a view to cover each aspect of cat ownership and with a holistic approach to caring for your feline companion.

Design: The frontend will be built in Flutter for cross platform support across both android and IOS. The backend will use a Node.js server to implement business logic for all the modules. It will also use a Hugging Face AI model for the chat bot and a Tensorflow machine learning model for breed detection. The database will be maintained on Firebase.

Purrfect Assistant combines technology with pet care knowledge, hoping to be a game changer for how cat owners interact and manage their pets' health and well being. The app is not only a tool, but also a companion to every cat owner, improving cats' quality of

life and improving the quality of life of pets and owners alike.

3.1 Architectural Design

3.1.1 High Level Architecture Diagram

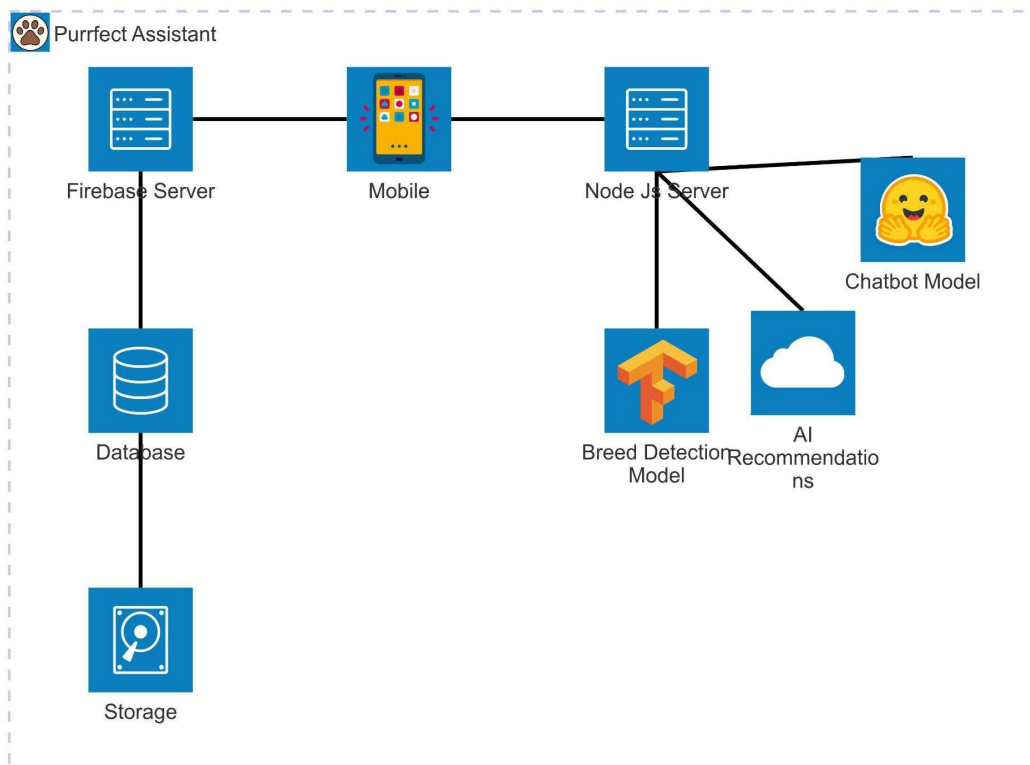


Figure 3.1: High Level Architecture Diagram

3.1.2 Detailed Architecture Diagram

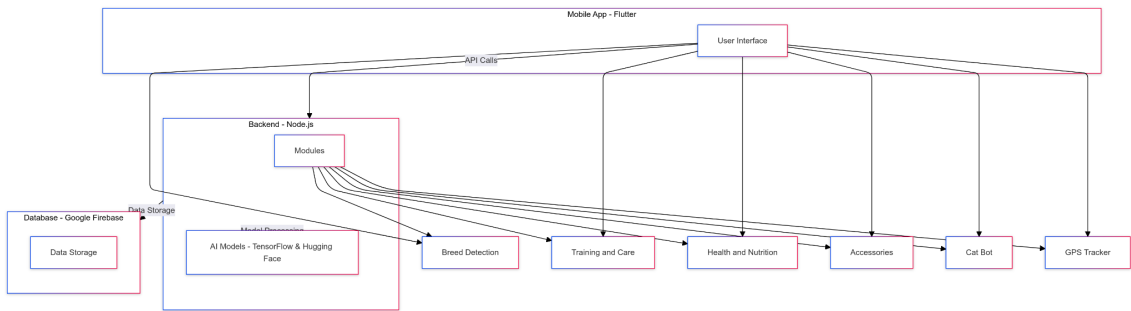


Figure 3.2: Detailed Architecture Diagram

3.2 Design Models

The project will follow Object Oriented Development Approach.

3.2.1 Activity Diagram

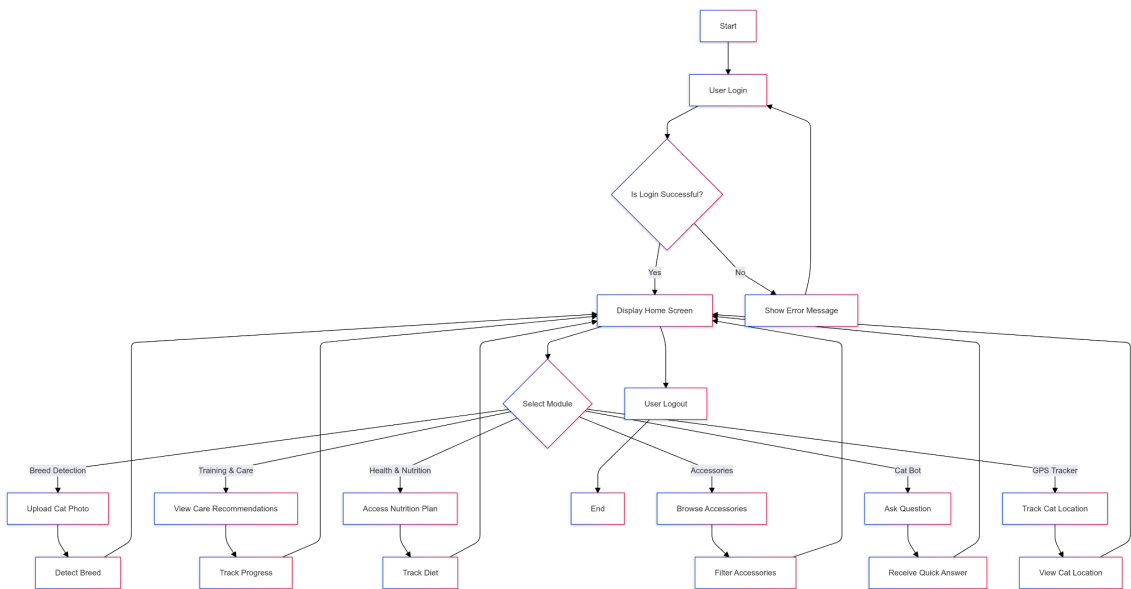


Figure 3.3: Activity Diagram

3.2.2 Class Diagram

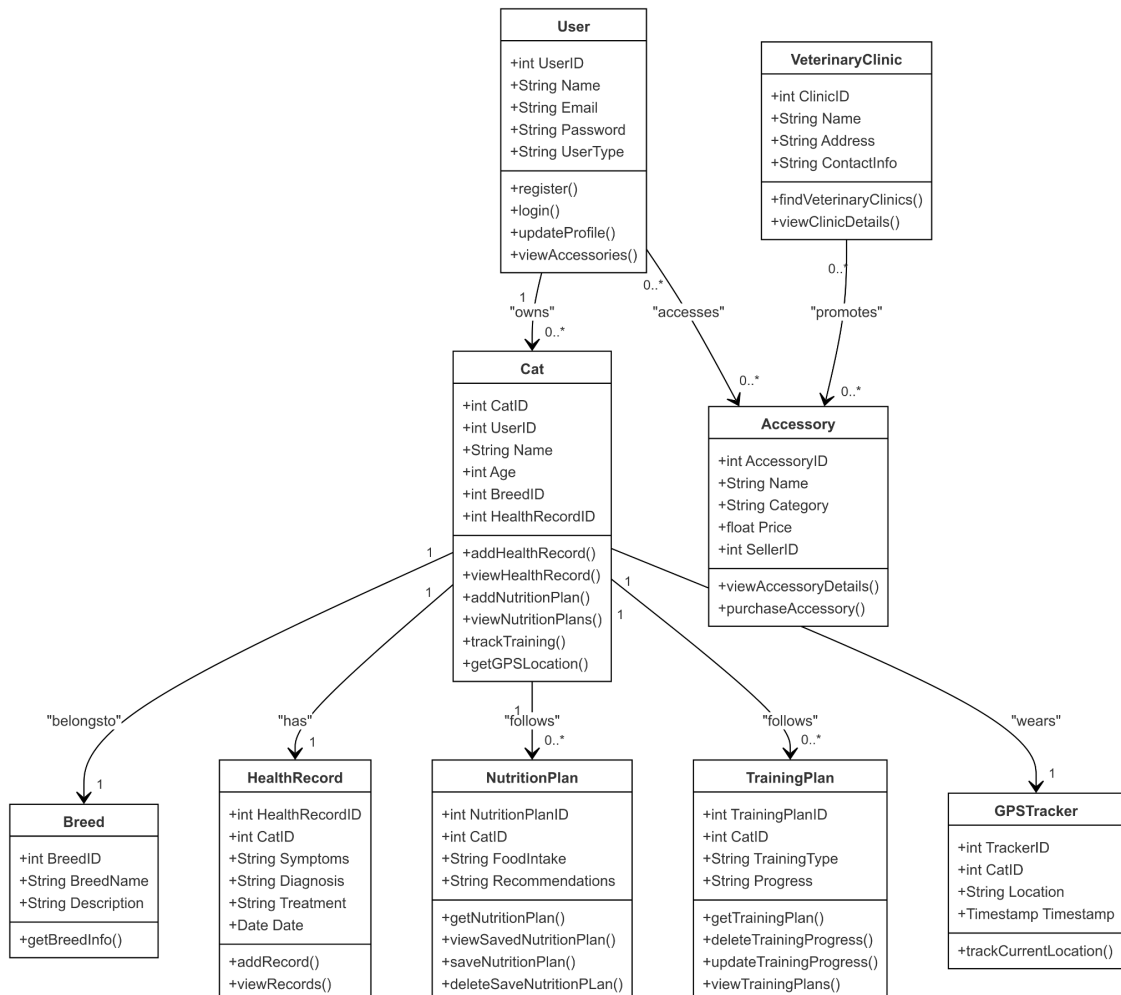


Figure 3.4: Class Diagram

3.2.3 Class-level Sequence Diagram

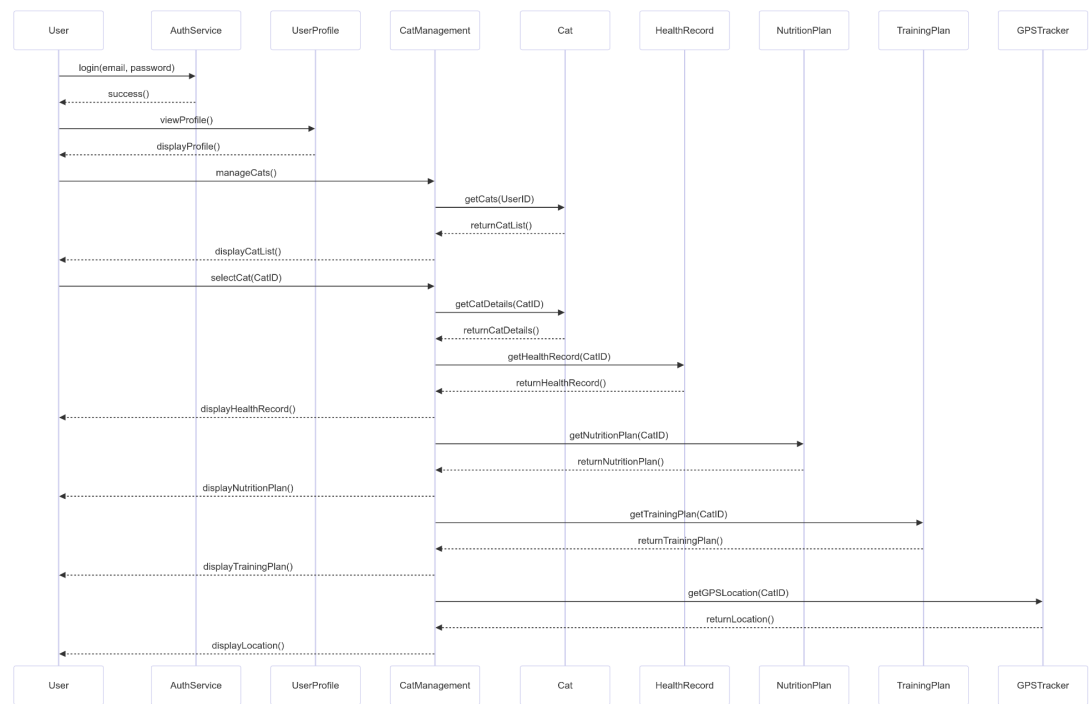


Figure 3.5: Class-level Sequence Diagram

3.2.4 State Transition Diagram

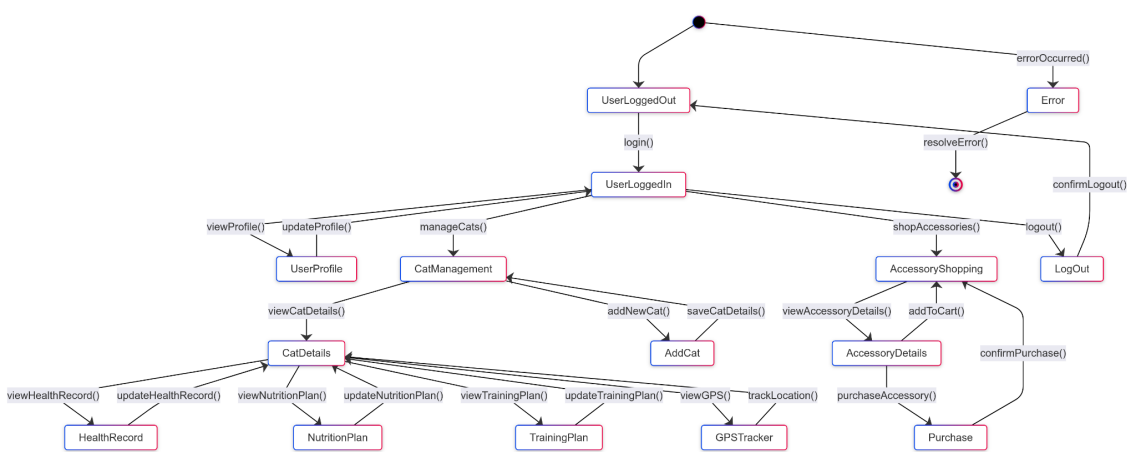


Figure 3.6: State Transition Diagram

3.3 Data Design

The Purrfect Assistant app takes its information domain from unstructured data to structured data, transforming it into well defined data models that can be stored and processed within **Firebase**, a cloud based NoSQL database. Using this approach, we can retrieve data smoothly, scale, and update real time, which improves the user experience and provides the means for managing cat care information in an effective manner.

The major data items that need to be stored will be:

1. User profiles
2. Cat profiles
3. Nutrition plans
4. Symptom analysis reports
5. Chat bot interactions

Firebase's real time database is mostly used for data processing inside the app. The app processes and stores the data corresponding to when users interact with the app, for example when uploading images for breed detection or logging symptoms. With Firebase, it is real time features that make it possible to get the latest information instantly across devices.

Bibliography